

Python Programming Assignments

Section 1: For Loop Questions

Object :1 [from w3resource.com]

Write a Python program to find those numbers which are divisible by 7 and multiple of 5, between 1500 and 2700 (both included).

Output: [1505, 1540, 1575, 1610, ..., 2695]

Object: 2

Write a program that prints Factorial of a user given integer without using `math.factorial()` function.

Example: Enter 5 → Factorial of 5 = 120

Object: 3 [from w3resource.com (for + list)]

Write a Python program that prints each item and its corresponding type from the following list.

Sample List: `datalist = [1452, 11.23, 1+2j, True, 'w3resource', (0, -1), [5, 12], {"class":'V', "section":'A'}]`

Output (sample): Value: 1452 | Type: int

Object : 4 [from w3resource.com]

Write a Python program to find numbers between 100 and 400 (both included) where each digit of a number is an even number. The numbers obtained should be printed in a comma-separated sequence.

Output: 200,202,204,206,208,220,222, ... ,488

Object: 5

Write a program that takes input an integer number and prints all of its Divisors.

Example: Enter 100 → Divisors: [1, 2, 4, 5, 10, 20, 25, 50, 100]

Object: 6

Write a program that takes input two integer numbers and prints all the common divisors.

Example 1: Numbers 100 and 80 → common divisors are 1, 2, 5, 20

Example 2: Numbers 72 and 90 → common divisors are 1, 2, 3, 6, 18

Test 1: 100 & 80 → [1, 2, 5, 20] | Test 2: 72 & 90 → [1, 2, 3, 6, 18]

Object: 7

Extend the logic of last object: write a program that takes input two integer numbers and prints the Highest Common Factor (also called Greatest Common Divisor).

Example 1: Numbers 100 and 80 → HCF = 20

Example 2: Numbers 72 and 90 → HCF = 18

Test 1: 100 & 80 → HCF = 20 | Test 2: 72 & 90 → HCF = 18

Section 2: For Loop Questions

Object: 1 [Function — Area of Rectangle]

Write a function that takes length and width as parameters and returns the area of a rectangle.

Object: 2 [Function — Check Even or Odd]

Write a function that takes an integer and returns whether it is Even or Odd.

Object : 3 [Function — Reverse a String]

Write a function that accepts a string and returns its reverse.

Object: 4 [Function — Sum of Digits]

Write a function that takes an integer and returns the sum of all its digits.

Object: 5 [Function — Prime Check]

Write a function that takes a number and returns True if it is prime, otherwise False.

Object : 6 [Function — Fibonacci Series]

Write a function that takes n and prints the first n Fibonacci numbers.

Example: $n=8 \rightarrow 0\ 1\ 1\ 2\ 3\ 5\ 8\ 13$

Object : 7 [Function — Count Vowels]

Write a function that accepts a string and returns the count of vowels in it.

Object : 8 [Function — Power without **]

Write a function that computes $\text{base}^{\text{exponent}}$ without using the `**` operator or `pow()`.

End of Assignments